

NOVA: MOCK UNIVERSITY WEBSITE

**REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE
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**FIVE YEARS INTEGRATED MASTERES (B.Sc. & M.Sc.)
IN
SOFTWARE DEVELOPMENT**

By

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DECLARATION

We, Gargee Kakaty and Kaustuva Kashyap, Roll Nos: IS-241-844-0013 (Gargee Kakaty) and IS-241-844-0022 (Kaustuva Kashyap), students of Five Years Integrated Masters (B.Sc. & M.Sc.) Program in Software Development of the department of Information Technology, Gauhati University hereby declare that we have compiled this report reflecting all our work during the semester long full time project as part of our curriculum.

We declare that we have included the descriptions etc. of our project work, and nothing has been copied/replicated from other's work. The facts, figures, analysis, results, claims etc. depicted in our report are all related to our full-time project work.

We also declare that the same report or any substantial portion of this report has not been submitted anywhere else as part of any requirements for any degree/diploma etc.

Gargee Kakaty
Branch:
Date:

Kaustuva Kashyap
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Date:



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Date:

CERTIFICATE

This is to certify that **Gargee Kakaty and Kaustuva Kashyap** bearing Roll Nos: *IS-241-844-0013 (Gargee Kakaty) and IS-241-844-0022 (Kaustuva Kashyap)* have carried out the project work of “NOVA” under my supervision and have compiled this report reflecting both the candidates work in the semester long project. The candidates did this project full time during the whole semester under my supervision, and the analysis, results, claims, etc. are all related to their studies and work during the semester.

I recommend submission of this project report as a part for partial fulfillment of the requirements for the degree of Five Years Integrated Masters (B.Sc. & M.Sc.) in Software Development of Gauhati University.

Bhabesh Kalita
Teaching Assistant

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Abstract

Academic institution websites increasingly serve as the primary medium through which universities communicate their identity, values, and offerings to a global audience. Yet many existing academic portals prioritize functional information delivery over user experience, leaving a significant gap between what these websites communicate and what users feel when navigating them. Nova was developed to address this gap by designing and building a GUI-centric mock university website that places user experience at the center of every decision.

The project was developed using exclusively core web technologies — HTML5, CSS3, and JavaScript — following a GUI-first methodology across four structured phases spanning design and wireframing, HTML/CSS implementation, JavaScript interactivity, and final testing and deployment. The resulting product comprises seventeen interconnected pages, a unified design system, responsive layouts, and fully functional client-side interactive components including a five-step application form, animated statistics counters, and dynamic navigation.

Nova successfully demonstrates that a framework-free, student-built frontend can produce a polished, professional, and emotionally resonant academic web experience. The application of consistent visual hierarchy, institutional color theory, accessible design practices, and systematic component architecture produced a cohesive digital identity capable of conveying institutional character and inspiring user engagement across all pages.

The findings suggest that rigorous application of GUI design principles, even within the constraints of native web technologies, is sufficient to bridge the gap between functional utility and meaningful user experience - offering a replicable model for how academic institutions can rethink their digital presence from the ground up.

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1. Introduction

1.1 Overview

NOVA is a comprehensive mock university website developed as a demonstration of contemporary web development practices, responsive design methodologies, and user-centered interface design principles. The project encompasses the complete development lifecycle, from requirements analysis and system design through implementation, usability testing, and evaluation.

The NOVA platform integrates multiple functional modules - including student admission portal, course directory, faculty information system, event management, and institutional news - within a cohesive, accessible, and responsive interface suitable for diverse user demographics and device types. [Figure 1 shows the architecture diagram of this project.]

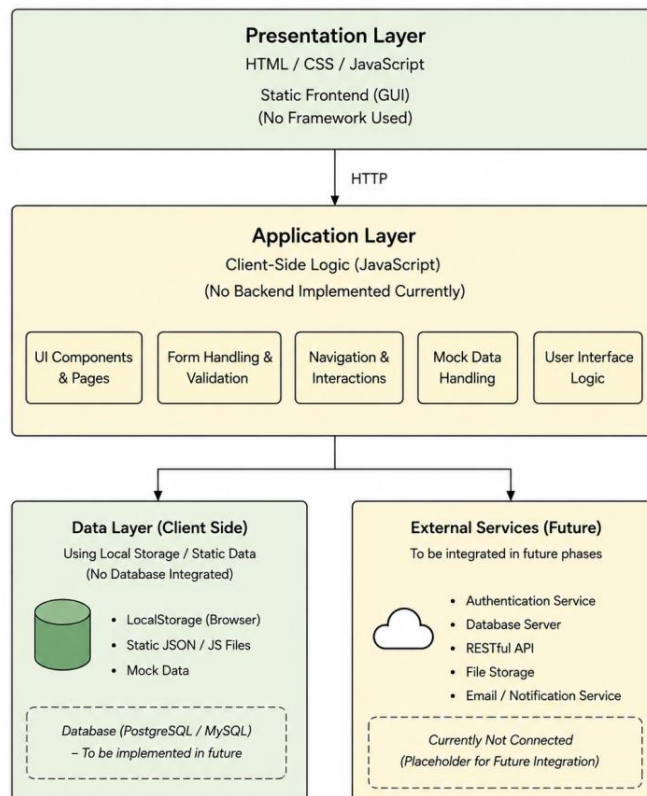


Figure 1 : Architecture Diagram

1.2 Problem Statement

Many existing university websites suffer from poor information architecture, inconsistent user experience across devices, accessibility issues, and outdated design paradigms. These deficiencies create barriers for prospective students, current students, faculty, and other stakeholders attempting to navigate institutional information. There is a demonstrable need for a modern university website that prioritizes user experience, incorporates accessibility standards, and employs responsive design principles to serve an increasingly mobile-first user base.

1.3 Objectives

The primary objectives of this project are:

1. To design a visually engaging, emotionally resonant, and user-friendly graphical user interface for an academic web platform, incorporating fundamental design principles such as alignment, contrast, spacing, typography hierarchy, and visual consistency across all pages and components.
2. To develop and integrate a functional backend infrastructure that supports dynamic data handling, form submissions, and content management — enabling the platform to operate beyond a static front-end system and simulate the behaviour of a production-grade institutional website.
3. To iteratively evaluate and refine the user interface and experience following backend integration, in accordance with the Agile Software Development Life Cycle — ensuring that each development sprint produces measurable improvements in usability, performance, and design coherence.

1.4 Scope

This project encompasses:

- 1. Frontend Development:** HTML5, CSS3, and JavaScript
- 2. Functional Modules:** Student admission portal, course catalog, faculty directory, event calendar viewer
- 3. Accessibility Features:** WCAG 2.1 Level AA compliance
- 4. Usability Evaluation:** User testing with feedback collection and analysis

2. Background

The digital presence of academic institutions has transformed significantly over the past two decades. What once served as simple online brochures have evolved into comprehensive, multi-functional portals that communicate an institution's identity, values, and culture to a globally distributed audience. For universities in particular, ***the official website functions as the primary point of first contact*** shaping perceptions of academic credibility, campus life, and institutional character long before a prospective student ever sets foot on campus. This reality places an unusually high design burden on academic websites, demanding that they simultaneously inform, inspire, and guide a diverse range of users through complex informational ecosystems.

A persistent challenge in academic web design is the tension between functional information delivery and emotional engagement. Many institutional websites *prioritize content density over user experience, resulting in interfaces that are technically complete but experientially flat*. Research and professional practice in the field of human-computer interaction consistently highlight that users form judgments about a digital platform within seconds of arrival, and that ***visual hierarchy, typographic clarity, and navigational consistency are decisive factors in whether a user chooses to engage further*** or abandon the site. The design of an academic portal must therefore address not only what information is communicated, but how that information is felt by the person encountering it.

Usability principles have long established that good interface design should ***minimize cognitive load, support recognition over recall, and maintain consistency across all pages and interactions***. In the context of a university website, these principles translate into practical decisions such as ***persistent navigation bars, breadcrumb indicators, clearly labeled dropdown menus, and structured multi-step forms*** that break complex processes into manageable stages. Each of these conventions reduces the mental effort required of users who may be navigating high-stakes processes such as submitting an application or locating admission deadlines.

Accessibility is another dimension that contemporary interface design can no longer be treated as optional. Standards governing web accessibility call for **sufficient color contrast, semantic markup, descriptive labeling of interactive elements**, and **designs that remain functional across assistive technologies**. Academic institutions, which serve students from diverse linguistic, cultural, and physical backgrounds, bear a particular responsibility to ensure their digital environments are inclusive and navigable by the broadest possible audience.

The evolution of responsive design has further reshaped expectations around academic web interfaces. With a significant and growing proportion of **users accessing institutional websites from mobile devices, layouts must adapt fluidly** across screen sizes without sacrificing hierarchy or readability. This requires a disciplined approach to CSS layout systems, thoughtful use of spacing and typography, and interface components that remain functional and legible regardless of viewport dimensions.

Finally, the question of visual identity deserves attention. Color, typography, and photographic language collectively construct a **brand-voice that either reinforces or undermines an institution's perceived standing**. Deep, authoritative tones paired with warm accent colors and serif display typography have long been associated with academic prestige, while clean sans-serif body text ensures that day-to-day informational content remains effortlessly readable. The convergence of these design decisions when applied consistently across every page of a multi-page platform produces a coherent digital identity that communicates institutional character with the same authority as the institution's physical spaces.

3. Methodology

Nova was developed following a GUI-first methodology - a deliberate inversion of the typical development sequence in which backend logic precedes visual design. In this project, all implementation decisions were anchored by interface and experience goals established at the outset, ensuring that technical execution consistently served design intent.

3.1 Agile SDLC Approach

The project adopted an Agile-inspired development model, dividing the workload into four iterative phases. Each phase produced a discrete, reviewable deliverable rather than deferring all evaluation to the end. This allowed design and implementation decisions to be validated incrementally with findings from one phase informing the next. Peer feedback and self-review checkpoints were embedded throughout to catch usability and consistency issues early.

Phase 1 — GUI Design and Wireframing

The project began with a structured design phase before a single line of code was written. Primary user flows were identified — browsing academic programs, navigating the admissions process, exploring campus life, and submitting the application form — and layout wireframes were sketched for page elements like CTA banner, Hero Banner, Nav-bar, etc. Visual identity decisions were made during this phase: the deep forest-green and warm gold color palette, the pairing of a serif display typeface with a geometric sans-serif for body text, and the recurring page template (hero banner → ticker bar → content sections → CTA banner → footer) were all established before implementation began. A CSS custom property sheet was drafted to encode all color tokens globally, ensuring the design system would remain consistent across the entire site.

Phase 2 — Core HTML/CSS Implementation

With the visual system defined, structural implementation began using semantic HTML5. Every page was built using appropriate semantic elements — `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>` — to ensure logical document structure and support for assistive technologies. CSS3 Flexbox and Grid were used throughout to construct responsive layouts, with each component designed to reflow gracefully from desktop to tablet to mobile viewports. The shared navbar and footer HTML was replicated identically across all 17 pages to enforce structural consistency,

and the global CSS variable sheet defined in Phase 1 was applied site-wide so that color and spacing changes could propagate instantly. Key pages completed in this phase included the Home page, About Us, Admission portal, Academic Calendar, and all four Program detail pages.

Phase 3 — JavaScript Interactivity

Once the static structure was in place, vanilla JavaScript was layered in to introduce client-side interactivity without introducing any external dependencies. The key components implemented in this phase were: a hamburger toggle for the mobile navigation menu; dropdown hover menus on the desktop navbar; an Intersection Observer-driven animated counter that counts up enrollment, faculty, alumni, and employment statistics when the relevant section scrolls into view; a horizontally scrolling marquee/ticker bar used as a visual separator across multiple pages; client-side form validation on both the Contact page form and the five-step Apply Now application form; and dynamic step-progression logic for the multi-step application flow, including visual state management for completed, active, and upcoming steps. All JavaScript was written in vanilla ES6+ and encapsulated to prevent scope conflicts between pages.

Phase 4 — Testing, Validation, and Deployment

The final phase focused on cross-device testing, accessibility validation, and live deployment. The site was tested across Chrome, Firefox, Safari, and Edge on desktop, and on mobile viewports via browser DevTools and physical devices. Each page was manually reviewed for layout breakpoints, navigation behavior, and interactive component reliability at varying screen widths. Accessibility compliance was checked against WCAG 2.1 guidelines, with particular attention to color contrast ratios between the dark navy backgrounds and body text, semantic label associations on form fields, and logical focus order for keyboard navigation. The completed site was deployed to GitHub Pages, providing a publicly accessible live URL

(gargee1989.github.io/Nova/) with no build tools or server configuration required. [Figure 2 shows all the stages of Agile methodology.]

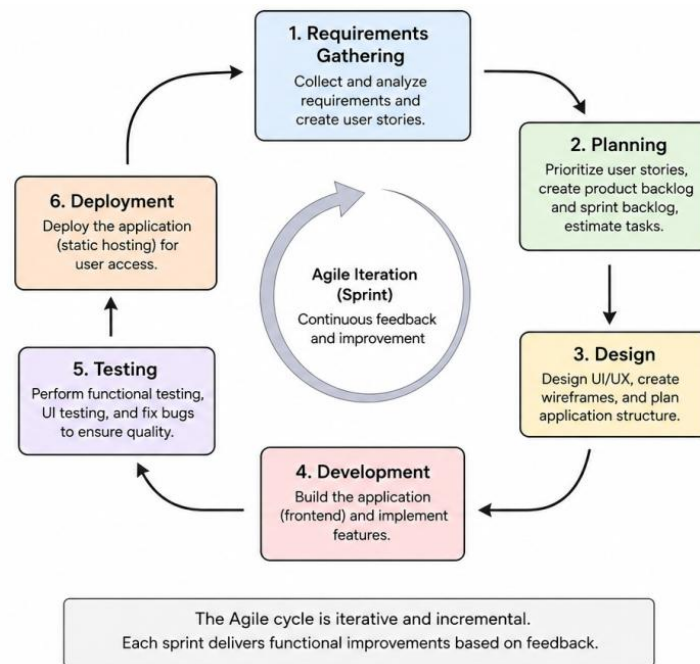


Figure 2: Agile model – SDLC

3.2 Tech Stack

A deliberate constraint of the project was to build entirely within core web technologies, with no external CSS frameworks (such as Bootstrap or Tailwind) and no JavaScript libraries (such as jQuery or React). This constraint was chosen to deepen fluency with the foundational technologies that all web frameworks are ultimately built upon and to demonstrate that a polished, production-grade frontend does not require third-party dependencies.

HTML5 provided the semantic document structure for all 17 pages + Apply Now page.

Semantic elements were used throughout to convey meaningful page structure to both browsers and assistive technologies, and to maintain a logical content hierarchy independent of visual styling.

CSS3 handles all visual styling, layout, and animation. Flexbox was used for one-dimensional alignment tasks such as navbar items, card rows, and footer columns, while CSS Grid was applied for two-dimensional layouts including the

programs grid, alumni card matrix, and department contact grid. CSS custom properties (variables) define the full color palette globally, enabling site-wide visual consistency and making design iteration efficient. Keyframe animations and CSS transitions provide hover state feedback and smooth visual interactions without JavaScript.

JavaScript was used exclusively for client-side interactivity. DOM manipulation powered the mobile navigation toggle and dropdown menus; the Intersection Observer API drove the scroll-triggered animated counters; form validation logic provided real-time user feedback on the Apply Now and Contact forms; and dynamic rendering managed the five-step application stepper's progress state. No server-side logic, build tooling, or package management was used at any point in the project.

GitHub Pages served as the deployment platform. The static nature of the project made GitHub Pages an ideal host — no build pipeline, no server configuration, and free public hosting directly from the project repository.

3.3 Team Roles and Division of Work

The project was developed collaboratively by two students of the 4th Semester Integrated Master's in Software Development program, with responsibilities divided according to complementary strengths.

Gargee Kakaty (IS-241-844-0013) served as Co-Developer and Scripter, taking primary responsibility for all JavaScript logic across the site. This included the mobile navigation hamburger toggle, DOM manipulation for dropdown menus, the Intersection Observer-driven animated statistics counters, client-side form validation for both the Contact and Apply Now pages. Gargee also handled mobile compatibility testing to ensure CSS and JavaScript-driven components functioned correctly across touch-based devices and narrow viewports. Readme and Project Report creation along with code organization, were handled by her to maintain a clean modular approach.

Kaustuva Kashyap (IS-241-844-0022) served as Co-Developer and Designer, responsible for the overall visual identity and CSS architecture of the site. This encompassed UI/UX design decisions, the definition of the global CSS color palette and typographic system, wireframing of page layouts, and the implementation of CSS3 Flexbox and Grid structures across all pages. Kaustuva led the visual consistency work — ensuring the shared navbar, footer, recurring section patterns, and component styles remained uniform site-wide. The Project Synopsis and Presentation were handled by her to have well-defined documentation for references.

Both team members participated in cross-device testing, peer review of each other's code, and the final accessibility validation pass prior to GitHub Pages deployment.

4. Implementation

4.1 Project Structure and File Architecture

The Nova website is organized as a multi-directory static file structure, with all pages, stylesheets, and scripts maintained in a single GitHub repository and deployed via GitHub Pages. The root directory holds the landing page (`index.html`) and the primary stylesheet, while inner pages are organized into subdirectories by category — `html/pages/` for general pages (About, Faculty, Events, Campus Life, Alumni), `html/academic/` for academic content (Academic Areas, Calendar), `html/programs/` for the four program detail pages, and `html/student services/` for Scholarships. The Apply Now, Admission, and Contact pages sit directly under `html/`. All images are stored in an `images/` folder referenced consistently across pages.

This structure reflects the site's navigational hierarchy and makes the relationship between pages and their content categories immediately legible to anyone reading the codebase.

4.2 Design System Implementation

Before individual pages were built, a global CSS custom property sheet was defined and linked across all pages. Every color used on the site is declared as a CSS variable at the `:root` level — `--navy` (`#1A221D`) for primary backgrounds, `--gold` (`#C9A84C`) for accent elements and CTAs, `--off-white` (`#F5F0EE`) for page backgrounds and body text on dark surfaces, and supporting tones for mid-panels, dividers, and muted text. This meant that any design iteration - adjusting a shade of green, shifting the gold accent - required a single change that propagated instantly across the entire site.

Typography followed the same systematic approach. A serif display typeface was applied to all hero headings and major section titles, while a geometric sans-serif handled all navigation labels, body copy, form fields, and UI text. A recurring typographic device - one word per major heading set in italics and gold - was implemented as a deliberate brand signature, appearing consistently across every page's hero banner.

4.3 Page-by-Page Implementation

Home Page (`index.html`) The homepage uses a split-panel hero layout: a dark left panel with the institution's name, tagline, and two CTA buttons, alongside a full-height campus photograph on the right. Below the hero, an animated statistics counter block displays four key figures (enrolled students, faculty members, alumni, and employment rate). These counters are driven by the Intersection Observer API — the count-up animation only triggers when the section enters the viewport, ensuring the effect is always seen rather than having already completed on page load. Subsequent sections present an About blurb, a programs grid, a campus life photo grid, a faculty carousel, student testimonials, and an upcoming events strip, all separated by the recurring gold ticker bar.

About Us (`html/pages/about.html`) The About page opens with a full-bleed hero banner carrying a breadcrumb trail (Home > About), followed by the gold ticker. A

year-founded counter block mirrors the homepage's animated counter implementation. The page then presents the university's founding story, a Mission and Core Values section rendered as an eight-card grid, a chronological milestones timeline, a campus facilities section, and a row of accreditation badges (QS World Rankings, NAAC, UN Academic Impact).

Admission Page (<html/admission.html>) The Admission page is structured as a single scrollable document with three anchored sections — Requirements, Admission Process, and Tuition Fees — navigable via pill-shaped anchor buttons near the top of the page. Requirements are presented as six numbered items; the process is rendered as a six-step horizontal timeline; tuition information is displayed in two structured tables. A FAQ section using an accordion interaction pattern (expand/collapse via JavaScript) closes the content before the CTA banner.

Academic Calendar (<html/academic/calendar.html>) The calendar uses a two-column layout: a main scrollable list of dated events (each entry showing a date block, event title and description, and a color-coded badge tag — Academic, Deadline, Exam, Break, Holiday, Ceremony) alongside a persistent right sidebar containing a "Coming Up" card with the next three upcoming events, a mini calendar widget, and a Download the Full Calendar CTA.

Scholarships (<html/student-services/scholarship.html>) Three scholarship cards are laid out in a CSS Grid row. Each card follows an identical component structure: a gold icon (trophy, globe, or chip), a badge tag (Merit Based, International, or Innovation), a heading and description, a bullet list of benefits with gold checkmark icons, and a full-width gold Apply Now button. The featured card is visually distinguished by a gold border. Below the cards, a three-step How to Apply process is rendered as a numbered flow.

Alumni Page (<html/pages/alumni.html>) The Alumni page features a four-column Notable Alumni card grid, where each card has a tall portrait image with a gradient overlay, the alumni name in bold white, their role in gold, and social icon

links. Below this, a two-column Featured Alumni spotlight presents a longer profile alongside a pull-quote. Animated counters (graduates, countries represented, employment rate, C-Suite leaders) appear in a stats ticker, implemented using the same Intersection Observer counter pattern as the homepage.

Apply Now (<html/applynow.html>) The most complex page on the site. A five-step horizontal progress stepper sits at the top of the form area, with step states managed in JavaScript — completed steps display a visual checkmark indicator; the active step is highlighted in gold, and future steps are rendered in muted grey. The form itself is divided into five panels: Personal Information, Academic Background, Program Selection, Documents and Essays, and Review and Submit. Only the active panel is visible at any time; JavaScript toggles panel visibility and updates the stepper on Next/Back navigation. A persistent right sidebar (~35% width) displays application deadlines with color-coded urgency badges and an application checklist, remaining visible regardless of which form step the user is on. A success confirmation screen replaces the form on mock submission.

Contact Page(<html/Contact.html>) The Contact page presents a contact information panel (address, office hours, social media links), a six-card department grid (each card showing the department name, phone number, email address, and operating hours), and a Visit Nova section listing upcoming guided campus tour slots with day, time, duration, and an accessibility note.

4.4 Shared Components

Certain UI components recur identically or with minor content variations across multiple pages and were implemented as consistent, reusable patterns.

Sticky Navbar: A dark navy navigation bar fixed to the top of every page. On desktop, it contains the university logo and wordmark on the left, three dropdown navigation groups (Pages, Academics, Student Services) in the center, and a gold Apply Now CTA button on the right. On mobile, the navigation links collapse behind a

hamburger icon, which toggles a full dropdown menu via JavaScript. Inner pages additionally display a slim utility bar above the main navbar showing contact phone, email, office hours, social media icons, and a language selector.

Dropdown Menus: Implemented in CSS using `:hover` states on the parent nav item to reveal absolutely positioned dark panels. The Academics dropdown uses internal sub-headings to group its nine links into three labelled categories (Academics, Admission, Programs), giving it more structure than a flat list.

Marquee / Ticker Bar: A horizontally scrolling strip of gold text on a dark background, implemented using a CSS `@keyframes` animation on a duplicated text node to produce a seamless infinite scroll. Each page uses unique ticker content — homepage shows live enrollment stats; the calendar page shows semester category labels; the alumni page shows graduate statistics.

Hero Banner: Every inner page opens with a full-width dark-overlay image banner. A breadcrumb path is positioned top-left, followed by a large serif h1 heading (with the italic gold accent word) and a subtitle paragraph in off-white. The overlay opacity is tuned per page to ensure sufficient contrast between the text and the background image.

CTA Banner: A full-width dark section reading "Begin Your Future at Nova University" with a sub-paragraph and two side-by-side buttons (gold-fill Apply Now and outlined Explore Programs). This section appears as the final content block before the footer on almost every page, implemented as a shared structural pattern across the HTML files.

Footer: A dark navy footer closes every page with the university logo and tagline on the left, three columns of quick links in the center, gold social media icons, and a copyright line. The footer HTML is replicated identically across all 17 pages.

4.5 Responsive Design Implementation

All layouts were built mobile-first, with CSS Grid and Flexbox handling reflow at standard breakpoints. Multi-column card grids (programs, alumni, departments) collapse to two columns on tablet and single-column on mobile. The sticky navbar transitions from its desktop dropdown layout to the hamburger toggle at the mobile breakpoint. The Apply Now sidebar stacks below the form on narrow viewports. Typography scales are adjusted at breakpoints to maintain readable hierarchy without overflow. The split-panel homepage hero collapses to a stacked single-column layout on mobile, with the campus photograph moving above the text panel.

4.6 Accessibility Implementation

Accessibility was treated as a design constraint rather than a post-hoc check. Semantic HTML5 elements (<header>, <nav>, <main>, <section>, <footer>) provide a logical document outline for screen readers. All form fields use explicit <label> elements with for attributes matching input IDs, along with placeholder text and descriptive helper text. Color contrast between the dark navy (#1A221D) backgrounds and off-white (#F5F0EE) text meets WCAG 2.1 AA contrast requirements. Interactive elements (buttons, links, form controls) have visible focus states for keyboard navigation. The multi-step form stepper uses numbered step indicators rather than relying solely on color to communicate progress state. The language selector (ENG) in the utility bar signals *readiness for multilingual expansion*. The Contact page explicitly notes that all campus tours are fully accessible and invites users to communicate accessibility requirements when booking.

[Figure 3 showcases the workflow of the entire website.]

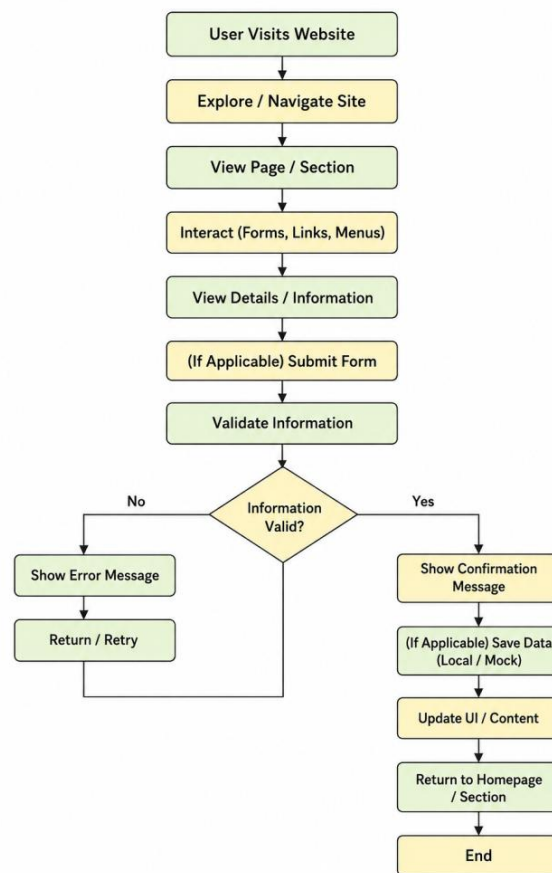


Figure 3: Workflow

5. Future Scope

The current iteration of the University of Nova website establishes a robust and visually refined front-end foundation. However, several technical enhancements can be incorporated in subsequent development phases to elevate the platform into a fully functional institutional web application.

1. **Backend Integration:** The website presently operates as a static front-end system with no server-side processing. A backend infrastructure — built using technologies such as Node.js with Express, or a cloud-based solution such as Firebase — would enable dynamic data handling, form submissions, and real-time content updates. Integration with a headless Content Management System (CMS) such as Sanity or Contentful would further allow non-technical

administrators to manage site content independently of the codebase. [Figure 4 represents the ER of the backend]

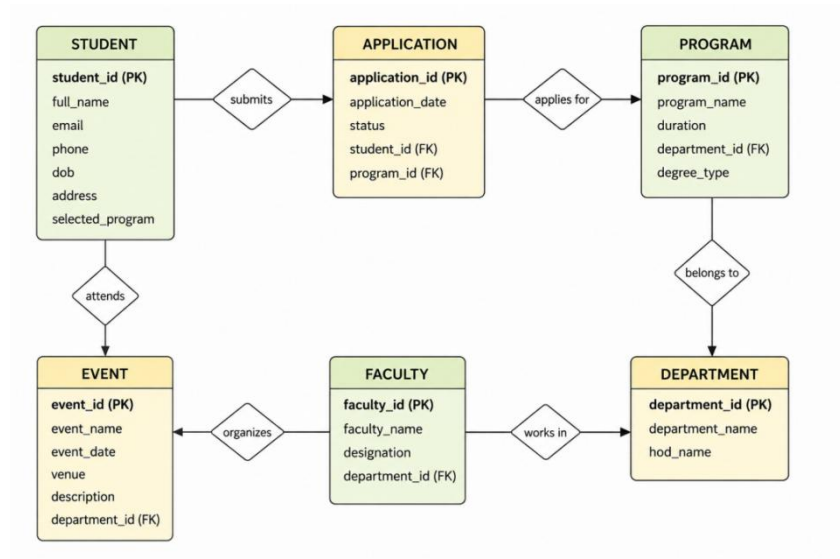


Figure 4: Entity-Relationship Diagram — Database Schema

2. **User Authentication & Portal System:** A secure authentication system can be implemented to support student and faculty login functionality. This would facilitate the development of a student portal encompassing features such as application status tracking, course enrollment, and fee payment dashboards — significantly extending the platform's utility beyond its current informational scope. [Figure 5 represents the use-case diagram of student & administrator]

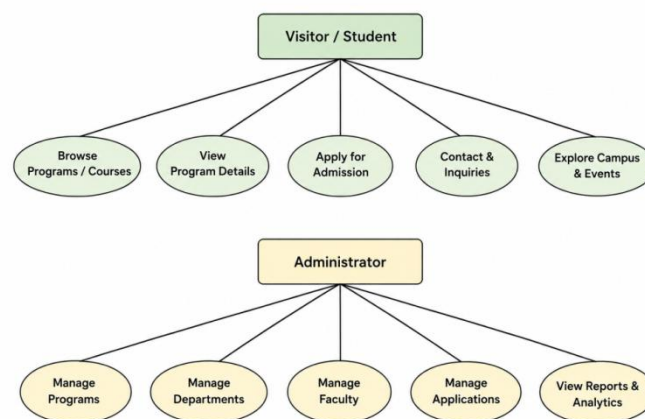


Figure 5: Use Case Diagram — Visitor/Student and Admin Functional Decomposition

3. **Database-Driven Content:** Faculty profiles, academic programs, events, and scholarship listings are currently hardcoded within the HTML structure. Migrating this content to a relational or document-based database would allow for scalable, query-driven rendering and eliminate the need for manual code edits when data changes.
4. **Search Functionality:** A site-wide search system, backed by a search index, would allow users to efficiently locate programs, faculty members, events, and academic resources — a feature essential to any institution of Nova's scale.
5. **Progressive Web Application (PWA) Conversion:** Converting the platform into a Progressive Web Application would enable offline accessibility, installability on mobile devices, and push notification support for time-sensitive communications such as application deadlines and academic calendar updates.
6. **API Integrations:** Third-party API connections — such as Google Calendar API for live event synchronisation, payment gateways for fee processing, and mapping APIs for campus navigation — would considerably enhance the functional depth of the platform. [Figure 6 represents the system module after backend integration & API integration]

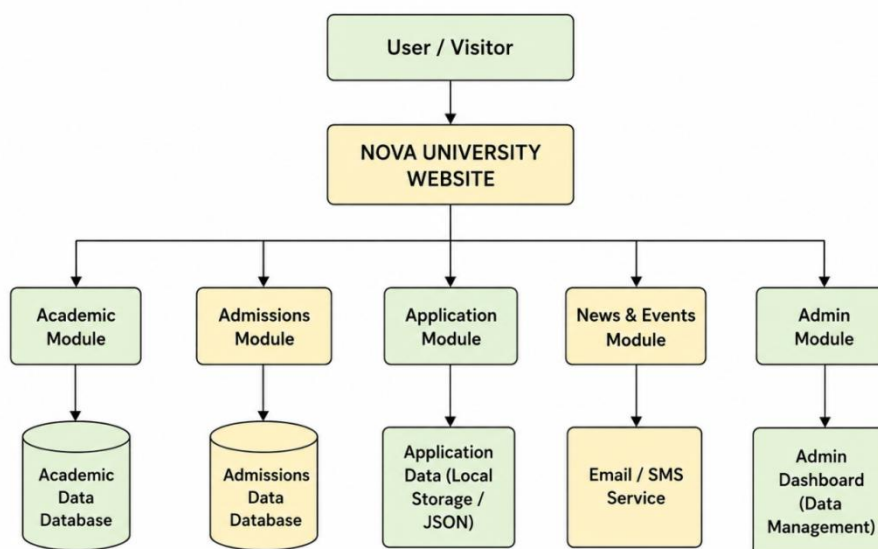


Figure 6: System Module Diagram

5. Conclusion

University websites occupy a uniquely important role in how academic institutions present themselves to the world, yet many fall short of their potential by prioritizing content delivery over user experience. Nova was developed as a direct response to this shortcoming - a demonstration that thoughtful, principles-driven GUI design can transform a digital academic portal into something that genuinely communicates the spirit and identity of an institution.

The development process followed a structured, four-phase GUI-first methodology, beginning with wireframing and visual identity, progressing through semantic HTML5 and CSS3 implementation, then JavaScript interactivity, and concluding with cross-device testing and deployment. This sequence proved effective, consistently reinforcing that design decisions made early in the process carried forward cleanly into implementation, reducing rework and producing a more coherent final product. The discipline of designing before building, and validating against established usability and accessibility standards throughout, was among the most valuable lessons the project yielded.

Nova contributes to a working demonstration that the gap between functional academic websites and genuinely engaging ones is not a technological barrier but a design one. It offers a replicable model, grounded in systematic color theory, typographic hierarchy, accessible component design, and consistent layout patterns that can inform how institutions, developers, and students alike approach academic web design.

Future iterations of this work could explore backend integration for live form submission and user authentication, a content management system for dynamic page updates, and more extensive accessibility auditing across assistive technologies and diverse user groups. Multilingual support, hinted by the language selector in the current design, also represents a meaningful avenue for expansion.

Nova ultimately demonstrates that thoughtful, principled design, even within significant technical constraints, is sufficient to create digital experiences that inform, engage, and inspire. That is, perhaps, the most important thing this project has to say.

6. References

- Wikipedia. (<https://en.wikipedia.org/wiki/Nova>)
- Wilbert O. Galitz, *The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques*, 3rd Edition, Wiley, 2007
- J. Nielsen, “10 Usability Heuristics for User Interface Design,” Nielsen Norman Group, 1995. (<https://www.nngroup.com/articles/ten-usability-heuristics/>)
- World Wide Web Consortium (W3C), “Web Content Accessibility Guidelines (WCAG) 2.1.” (<https://www.w3.org/TR/WCAG21/>)
- ThemeWant. (2024). *Unipix – University WordPress Theme* (<https://themewant.com/products/wordpress/unipix/>)
- Google, “Material Design — Design Principles.” (<https://m3.material.io/foundations>)
- Microsoft, “Fluent Design System.” (<https://fluent2.microsoft.design/>)